

Algebra: Simplifying Radical Expressions

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DIRECTIONS

Simplify each radical expression. Combine like radicals where applicable.

1 Simplify:

$$5\sqrt{7} - 2\sqrt{7}$$

Answer: _____

2 Simplify the radical:

$$\sqrt{200}$$

Answer: _____

3 Simplify:

$$2\sqrt{3} + 3\sqrt{12}$$

Answer: _____

4 Simplify the radical:

$$\sqrt{50}$$

Answer: _____

5 Simplify the radical:

$$\sqrt{98}$$

Answer: _____

6 Simplify the radical:

$$\sqrt{75}$$

Answer: _____

7 Simplify:

$$3\sqrt{2} + 5\sqrt{2}$$

Answer: _____

8 Simplify:

$$4\sqrt{5} + \sqrt{5}$$

Answer: _____

9 Simplify the radical:

$$\sqrt{180}$$

Answer: _____

10 Simplify the radical:

$$\sqrt{108}$$

Answer: _____



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Answer Key & Solutions

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TEACHER NOTES Factor out perfect squares first. Combine only like radicals (same index and radicand).

1 Simplify:

$$= 3\sqrt{7}$$
$$5\sqrt{7} - 2\sqrt{7}$$

2 Simplify the radical:

$$= 10\sqrt{2}$$
$$\sqrt{200}$$

3 Simplify:

$$= 8\sqrt{3}$$
$$2\sqrt{3} + 3\sqrt{12}$$

4 Simplify the radical:

$$= 5\sqrt{2}$$
$$\sqrt{50}$$

5 Simplify the radical:

$$= 7\sqrt{2}$$
$$\sqrt{98}$$

6 Simplify the radical:

$$= 5\sqrt{3}$$
$$\sqrt{75}$$

7 Simplify:

$$= 8\sqrt{2}$$
$$3\sqrt{2} + 5\sqrt{2}$$

8 Simplify:

$$= 5\sqrt{5}$$
$$4\sqrt{5} + \sqrt{5}$$

9 Simplify the radical:

$$= 6\sqrt{5}$$
$$\sqrt{180}$$

10 Simplify the radical:

$$= 6\sqrt{3}$$
$$\sqrt{108}$$

